

第一章 多元函数微分学

习题 1.0.1. 设可微函数 $z = z(x, y)$ 满足 $x^2 \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = 2z^2$, 又设 $u = x, v = \frac{1}{y} - \frac{1}{x}, w = \frac{1}{z} - \frac{1}{x}$, 则对于函数 $w = w(u, v)$, 求 $\left. \frac{\partial w}{\partial u} \right|_{\substack{u=2 \\ v=1}}$

解. 已知 $u = x, v = \frac{1}{y} - \frac{1}{x}, w = \frac{1}{z} - \frac{1}{x}$.

故

$$x = u, \quad \frac{1}{y} = v + \frac{1}{u}, \quad \frac{1}{z} = w + \frac{1}{u}$$

对 $w = w(u, v)$ 关于 u 求偏导:

$$\frac{\partial w}{\partial u} = \frac{\partial}{\partial u} \left(\frac{1}{z} - \frac{1}{u} \right) = -\frac{1}{z^2} \frac{\partial z}{\partial u} + \frac{1}{u^2}$$

其中 z 是 x, y 的函数, 而 x, y 又是 u, v 的函数. 由复合函数求导法则:

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u}$$

由 $x = u$ 知 $\frac{\partial x}{\partial u} = 1$. 由 $\frac{1}{y} = v + \frac{1}{u}$ 知 $-\frac{1}{y^2} \frac{\partial y}{\partial u} = -\frac{1}{u^2}$, 即 $\frac{\partial y}{\partial u} = \frac{y^2}{u^2} = \frac{y^2}{x^2}$.

代入得:

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \cdot 1 + \frac{\partial z}{\partial y} \cdot \frac{y^2}{x^2} = \frac{1}{x^2} \left(x^2 \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} \right)$$

已知原方程 $x^2 \frac{\partial z}{\partial x} + y^2 \frac{\partial z}{\partial y} = 2z^2$, 则:

$$\frac{\partial z}{\partial u} = \frac{2z^2}{x^2}$$

代入 $\frac{\partial w}{\partial u}$ 的表达式:

$$\frac{\partial w}{\partial u} = -\frac{1}{z^2} \cdot \frac{2z^2}{x^2} + \frac{1}{x^2} = -\frac{2}{x^2} + \frac{1}{x^2} = -\frac{1}{x^2}$$

当 $u = 2, v = 1$ 时, $x = u = 2$, 故:

$$\left. \frac{\partial w}{\partial u} \right|_{\substack{u=2 \\ v=1}} = -\frac{1}{2^2} = -\frac{1}{4}$$

□

习题 1.0.2. 设 $z = f(x, y)$ 是区域 $D = \{(x, y) | 0 \leq x \leq 1, 0 \leq y \leq 1\}$ 上连续可微的函数,

且满足 $f(0, 0) = 0$, 且 $dz|_{(0,0)} = 3dx + 2dy$, 求极限 $\lim_{x \rightarrow 0} \frac{\int_0^{x^2} dt \int_x^{\sqrt{t}} f(t, u) du}{1 - \sqrt[4]{1 - x^4}}$

解. 设极限为 L . 首先处理分母: 当 $x \rightarrow 0$ 时, 使用等价无穷小 $(1+x)^\alpha - 1 \sim \alpha x$, 有:

$$1 - \sqrt[4]{1 - x^4} = 1 - (1 - x^4)^{1/4} \sim -\left[\frac{1}{4}(-x^4)\right] = \frac{1}{4}x^4$$

于是:

$$L = \lim_{x \rightarrow 0} \frac{\int_0^{x^2} dt \int_x^{\sqrt{t}} f(t, u) du}{\frac{1}{4}x^4}$$

接下来处理分子, 对分子先进行交换积分次序:

$$\int_0^{x^2} dt \int_x^{\sqrt{t}} f(t, u) du = - \int_0^x du \int_0^{u^2} f(t, u) dt$$

容易发现整个分式是 $\frac{0}{0}$ 型的, 因此可以使用洛必达法则. 对分子分母求导:

$$L = \lim_{x \rightarrow 0^+} \frac{- \int_0^x du \int_0^{u^2} f(t, u) dt}{\frac{1}{4}x^4} = - \lim_{x \rightarrow 0^+} \frac{\int_0^{x^2} f(t, x) dt}{x^3}$$

容易发现分子分母又是 $\frac{0}{0}$ 型的, 但如果继续使用洛必达法则, 分子是变限积分, 求导后积分仍然存在, 不如使用定积分中值定理.

由定积分中值定理, 存在 $\xi \in (0, x^2)$ 使得:

$$\int_0^{x^2} f(t, x) dt = f(\xi, x) \cdot x^2$$

因此:

$$L = - \lim_{x \rightarrow 0^+} \frac{f(\xi, x) \cdot x^2}{x^3} = - \lim_{x \rightarrow 0^+} \frac{f(\xi, x)}{x}$$

当 $x \rightarrow 0^+$ 时, $\xi \rightarrow 0^+$, 因此:

$$f(\xi, x) = f(0, 0) + \frac{\partial f}{\partial x}(0, 0) \cdot \xi + \frac{\partial f}{\partial y}(0, 0) \cdot x + o(\sqrt{\xi^2 + x^2})$$

代入得:

$$L = - \lim_{x \rightarrow 0^+} \frac{f(0,0) + 3\xi + 2x + o(\sqrt{\xi^2 + x^2})}{x} = -2 + 3 \lim_{x \rightarrow 0^+} \frac{\xi}{x} + \lim_{x \rightarrow 0^+} \frac{o(\sqrt{\xi^2 + x^2})}{x}$$

由于 $0 < \xi < x^2$, 所以 $0 < \frac{\xi}{x} < x$, 因此 $\lim_{x \rightarrow 0^+} \frac{\xi}{x} = 0$. 而

$$\lim_{x \rightarrow 0^+} \frac{\sqrt{\xi^2 + x^2}}{x} = \lim_{x \rightarrow 0^+} \frac{o(\sqrt{\xi^2 + x^2})}{\sqrt{\xi^2 + x^2}} \cdot \frac{\sqrt{\xi^2 + x^2}}{x} = 0$$

因此:

$$L = -2$$

□

注. 关于多重变限积分的求导, 事实上是莱布尼茨积分法则的推广, 具体来说, 如果 $F(x) = \int_{a(x)}^{b(x)} f(t, x) dt$, 则:

$$F'(x) = f(b(x), x) \cdot b'(x) - f(a(x), x) \cdot a'(x) + \int_{a(x)}^{b(x)} \frac{\partial f(t, x)}{\partial x} dt$$

在本题中, 分子中的积分是一个二重变限积分, 因此需要对内外两层积分分别使用莱布尼茨法则进行求导.